The Volume Inflection Model

Findings and limits

Three external lengths can grow while a fourth, circular direction contracts. For one tuned solution of the five-dimensional Einstein equations, the total spatial volume falls through an instant of zero velocity and zero acceleration. Its maximum occurs earlier. This is a result about a specified matter law and initial conditions. No cosmological observation has identified the instant.

What the equations require

The spatial volume is the product of the three equal external lengths and the circle circumference. At the event, the circle contracts at three times the fractional growth rate of one external length. The flat-space gravitational constraint then requires negative total energy density. The prescribed inverse-fifth-power stress gives the falling inflection. A periodic scalar treated with all four finite periods misses its exact cubic coefficient. At equal periods it gives a flat-topped maximum instead. The distinction is proved on pages 3–4.

Order of events in the calculated history

Total spatial volume reaches its maximum	−0.431
Circle circumference reaches its maximum	−0.173
Circle size relative to one external length reaches its maximum	−0.157
External lengths reach their minimum	−0.127
Total spatial volume falls through the stationary inflection	0

Times are rounded and measured in units of the external expansion time at the marked instant. Continuing the assumed stress gives curvature singularities at both ends in finite proper time. The slowly varying circle approximation fails before those limits.

Illustrated account Technical paper and proofs

What the observations could establish

Five exact volumes at the specified external lengths force a bend in each direction. The four successive losses are about 23.90, 3.18, 2.84 and 18.75 parts per million. Even so, any finite selection of length-indexed readings from this curve admits an analytic pressure law for which the volume slope is strictly negative throughout the tested interval. The construction does not fix elapsed times between readings.

What a spectral line can tell us

In the worked example, one circle-sensitive line arrives at 75.3% of the frequency of the same local transition. With a stable emission law this identifies the change in the ratio of circle size to external length. A second stable standard would recover the 0.46% fall in volume. An unnoticed change of about 0.115% in the ordinary reference could conceal that fall.

What remains physically unresolved

The finite static scalar benchmark already fails the cubic test at every finite aspect ratio. Its instantaneous use on a moving torus is an extra assumption. A physical realization needs a field state, its time-dependent stress, and an emitter with characterized transitions. A four-dimensional scalar with matched couplings reproduces the listed frequencies. Cassini radio tracking excludes the particular present-day, long-range, unscreened, universally coupled scalar interpretation. It does not date the proposed event.

Cassini measurement

Vacuum stress in compact cosmologies

Verification

The published nine-page illustrated account and fifteen-page paper were compared page by page with the checked copies, including the contents links and bookmarks. All four embedded calculations reproduced identical outputs in ordinary Python and its two optimized modes. A separate proper-time integration recovered the turning order above. Retain the prescribed solution as a mathematical example. A physical interpretation now has a specific source test to pass (page 4).

A direct test of the proposed stress

Keep the flat homogeneous metric, Einstein equations, positive constant ρ_Λ and three equal external periods a. Replace only the negative density and its pressures. Use one massless periodic real scalar on a static rectangular torus, subtract the decompactified vacuum, and insert the resulting stress instantaneously in the evolution equations. This replacement tests a specific matter prescription. It provides no derivation on an evolving torus.

All finite periods change the cubic coefficient

Write $L = 2\pi b$ for the circumference. The static topological density is

$$f = -\frac{3}{8\pi^2} \sum_{(n,m)\neq(0,0)} (a^2|n|^2 + L^2 m^2)^{-5/2}, \qquad (n,m) \in \mathbb{Z}^3 \times \mathbb{Z}.$$

Keeping $n = 0$ alone recovers the circle-only density in the paper. The full series and its log-length derivatives converge locally uniformly, by comparison to the four-dimensional sum of inverse fifth powers. Virtual work gives $p_x = -f - af_a/3$, $p_y = -f - Lf_L$. Homogeneity, $af_a + Lf_L = -5f$, makes this component traceless. Consequently $\ddot{V}/V = \kappa(8\rho_\Lambda/3 + f)$, where $V = a^3 L$.

At a tuned event with $H = \dot{a}/a > 0$ and $\dot{V} = \ddot{V} = 0$, $K = \dot{L}/L = -3H$, $f = -8\rho_\Lambda/3$, and $\kappa f/H^2 = -48/5$. For any chosen aspect ratio the overall length can be scaled to satisfy this tuning. With $x = \log(a/a_*)$ and $R = \log(V/V_*)$, differentiation at the event yields

$$c_3 = \tfrac{1}{6}R'''(0) = \frac{\kappa(af_a - 3Lf_L)}{6H^2} = -24 + 32\left\langle \frac{a^2|n|^2}{a^2|n|^2 + L^2m^2} \right\rangle.$$

The brackets denote an average with normalized positive weights proportional to $(a^2|n|^2 + L^2m^2)^{-5/2}$. Both pure-circle and pure-external windings have positive weight. Hence $-24 < c_3 < 8$ for every finite positive a, L. Retuning ρ_Λ, the length scale or the expanding event data cannot give $c_3 = -24$ in this prescription. At fixed L the nonzero external windings vanish from the average as $a/L \to \infty$, by dominated convergence. The mismatch therefore gives no uniform finite-precision separation.

Numerical illustration

a/L	1/2	1	2	5	10
c_3	4.80846	0	−13.77885	−23.57191	−23.97282

These rounded quadratures illustrate the exact inequality. For example, an aspect ratio of ten still permits a falling stationary inflection, but with a different coefficient. The zero at equal periods follows exactly from permutation symmetry.

At equal periods the event changes type

Fix the spatial volume and write $a = \ell e^{s}$, $L = \ell e^{-3s}$. Set $Z(s) = \sum'(e^{2s}|n|^2 + e^{-6s}m^2)^{-5/2}$, so f is a negative constant times $\ell^{-5}Z(s)$. The Mellin representation integrates the product

$$\vartheta(te^{2s})^3\vartheta(te^{-6s}) - 1, \qquad \vartheta(t) = \sum_{k\in\mathbb{Z}} e^{-tk^2},$$

with positive weight $t^{3/2}/\Gamma(5/2)$. The small-t term from Poisson summation depends only on volume. The shape derivatives of the integrand decay exponentially there after removal of that term, and at infinity.

Sign of the shape response

For $g(z) = \log\vartheta(e^{z})$, one has $g'' > 0$. To see this, take $t = e^{z}$, and give integer k weights proportional to e^{-tk^2}. Then $g'' = t^2\,\mathrm{Var}(k^2) - t\,\mathbb{E}[k^2]$. For $t \geq \pi$, $\mathbb{E}[k^2] \leq 2\sum_{k\geq 1} k^2 e^{-3k} < 1/2$. Since $k^4 \geq k^2$, the variance is at least $\mathbb{E}[k^2](1-\mathbb{E}[k^2])$, which proves strict positivity. For $t < \pi$ use $\vartheta(t) = \sqrt{\pi/t}\,\vartheta(\pi^2/t)$; its log prefactor is affine in z. This also gives positivity on the remaining range.

The logarithm of the theta product has derivative $6[g'(\log t + 2s) - g'(\log t - 6s)]$, with the sign of s. Its second derivative at zero is $48g''(\log t) > 0$. Hence $Z'(0) = 0$, $Z''(0) > 0$, and Z' has the sign of s. At the tuned event $c_3 = -(8/5)Z'/Z$. A longer external period gives a falling inflection; a shorter one gives a rising inflection.

At equal periods the cubic vanishes. The actual path has $\log\ell = \frac{1}{4}\log V$ stationary through second order, and $\dot{s} = H_*$ at the event. Therefore $\ddot{f}_* = H_*^2 f_{ss}$ there. Differentiating the volume acceleration once more gives

$$R(x) = c_4 x^4 + O(x^5), \qquad c_4 = -\tfrac{2}{5}Z''(0)/Z(0) < 0.$$

This is a strict local maximum. Quadrature gives $c_4 \simeq -10.58693$; the sign follows from the proof.

What would move the model forward

Specify a quantum state, field content, boundary conditions and renormalization prescription on the evolving torus. Compute the stress and its time derivative with errors small enough to test three requirements at one event. They are $\kappa\rho = -6H_*^2$, $J = 0$, and $\kappa\dot{J} = -432H_*^3$, with $J = 4\rho - 3p_x - p_y$ for the total stress. The last requirement distinguishes the cubic prediction even after initial data have been retuned. Until this test is met, the singular continuation and the observational examples remain predictions of the prescribed equations.

Ambjørn and Wolfram, finite-volume vacuum energy. Faulhuber and Steinerberger, theta-function convexity. Saharian and Mkhitaryan, state and geometry dependence of vacuum stress. The attached Python file checks the quadratures by differentiation and symmetric lattice cutoffs.